

Regulatory genome annotation of 33 insect species

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Abstract

Annotation of newly-sequenced genomes frequently includes genes, but rarely covers important non-coding genomic features such as the *cis*-regulatory modules—e.g., enhancers and silencers—that regulate gene expression. Here, we begin to remedy this situation by developing a workflow for rapid initial annotation of insect regulatory sequences, and provide a searchable database resource with enhancer predictions for 33 genomes. Using our previously-developed SCRMshaw computational enhancer prediction method, we predict over 2.8 million regulatory sequences along with the tissues where they are expected to be active, in a set of insect species ranging over 360 million years of evolution. Extensive analysis and validation of the data provides several lines of evidence suggesting that we achieve a high true-positive rate for enhancer prediction. One, we show that our predictions target specific loci, rather than random genomic locations. Two, we predict enhancers in orthologous loci across a diverged set of species to a significantly higher degree than random expectation would allow. Three, we demonstrate that our predictions are highly enriched for regions of accessible chromatin. Four, we achieve a validation rate in excess of 70% using *in vivo* reporter gene assays. As we continue to annotate both new tissues and new species, our regulatory annotation resource will provide a rich source of data for the research community and will have utility for both small-scale (single gene, single species) and large-scale (many genes, many species) studies of gene regulation. In particular, the ability to search for functionally-related regulatory elements in orthologous loci should greatly facilitate studies of enhancer evolution even among distantly related species.

eLife assessment

This **important** manuscript uses a machine-learning approach to predict and annotate cis-regulatory elements across insect genomes, helping to address a much-needed gap in comparative genomics. This method does not rely on sequence alignments, thereby allowing functional genomics studies of more distant species, including emerging model organisms. There are nuanced views on the strength of the evidence from the predictions: the pipeline appears to be based on **solid** evidence, but the methods could be better described. We suggest the manuscript would be much more robust if the code used was accessible for review and validated further.

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Introduction

The past two decades have witnessed an explosive rise in sequenced metazoan genomes, from a mere handful in the first few years of the century to over 8,000 today (ref. 1; accessed 16 January 2024). This impressive statistic, however, masks the reality that these genomes exist in various stages of completion. Fewer than 30% of these genomes are assembled at the chromosome level, and only 28% of those have a comprehensive annotation (1). Moreover, almost none of the genome annotations include regulatory sequences (also referred to as cis-regulatory modules, or CRMs) such as enhancers and silencers. This is unfortunate, as CRMs comprise a significant percentage of the genome, and knowledge of these sequences is expected to have value comparable to that of knowing the protein-coding genes. Characterizing CRMs is critical for understanding mechanisms of gene regulation and the organization of gene regulatory networks. Moreover, the role of regulatory mutations is increasingly recognized as a driver of both evolution and disease (2–6).

One reason for the overall dearth of regulatory annotations is that, historically, large-scale CRM discovery has been difficult and both resource and labor intensive (7). For much of the last four decades, CRMs could only be identified through painstaking, low-throughput experimental assays. Although in recent years high-throughput empirical and computational CRM discovery methods have been developed, the various different methods frequently show limited agreement (8–10), with the result that comprehensive CRM annotation across all cell types and life-cycle stages has remained a challenge for all but the most exhaustively studied model organisms. The problem is particularly acute for the insects. Insects represent a species-rich class—they constitute somewhere between 65%–90% of all animal species (11, 12)—and have major impacts on human health and agriculture. The early radiation of the insects, coupled with typically short generation times, means that most of the relevant biomedically and agriculturally important species share little non-coding sequence conservation with each other or with the principal insect model species, *Drosophila melanogaster*. Thus, common sequence-homology based CRM discovery approaches are of little use in providing regulatory insights into these species, and knowledge transfers poorly from one species to another. Furthermore, many insects have a complex and varied life cycle, making it particularly important—yet onerous—to assay for CRM function at multiple stages.

We previously developed a powerful computational method, SCRMshaw (“Supervised Cis-Regulatory Module prediction”), for accurate prediction of CRMs, particularly enhancers (13–15). (Although we expect SCRMshaw to be adept at finding multiple CRM types, our validation

efforts to date have focused solely on enhancers.) SCRMshaw requires only a sequenced genome and a “training set” of some 15-30 known enhancers that regulate a common pattern of gene expression (e.g., midgut expression) and relies on the idea that enhancers with similar function will have similar sequence characteristics, not possible to detect by eye or by traditional alignment methods, but identifiable using machine-learning. Although not universally true, this assumption is robust enough to allow effective enhancer discovery without requiring knowledge of transcription factor binding sites or of the expression patterns of the genes being regulated.

Importantly, we have shown that we can leverage the wealth of existing *D. melanogaster* enhancer data (16) to train models for *cross-species* supervised enhancer discovery in diverged (160-345 million years (MY)) insect species—including flies, mosquitoes, beetles, bees, and wasps—despite a virtually complete lack of observable alignment at the non-coding sequence level (17–20).

Here, we use SCRMshaw to undertake an initial regulatory annotation of 33 individual insect genomes, using a collection of 48 training sets composed of experimentally validated *D. melanogaster* enhancers. These species are spread across five orders spanning over 360 MY of evolution, and represent roughly 10% of annotated insect species with scaffold-level or better assembly. Annotated predicted enhancers are provided in a searchable database that allows querying by species, tissue/cell type, or potential target gene. A series of simulations as well as *in silico* and *in vivo* validation experiments demonstrate the effectiveness of our approach and place an upper bound on false positive prediction rates. Our results represent the first release of a rich insect regulatory genome annotation resource, which will continue to grow and annotate insect regulatory genomes in parallel with the sequencing of new insect genomes.

Results

We previously demonstrated that SCRMshaw is remarkably effective at predicting enhancers across the entire ~345 MY range of the holometabolous insects, using training data derived solely from *D. melanogaster* (17, 19). SCRMshaw (Fig. 1A) uses training sets composed of known enhancers defined by a common functional characterization (e.g. “nervous system,” “wing disc”) to build a statistical model that captures their short DNA subsequence (kmer) count distribution. This kmer distribution is then compared to that of a set of non-enhancer “background” sequences in a machine-learning framework. The kmers likely serve as proxies for the unknown transcription factor binding sites, but these sites themselves, even when known, are not explicitly used by the algorithm. The trained model is then used to score overlapping sequence windows in the genome, and the highest-scoring windows are output as predicted enhancers (13, 14, 21). When searching the genomes of the mosquitoes *Anopheles gambiae* and *Aedes aegypti*, the red flour beetle *Tribolium castaneum*, the honey bee *Apis mellifera*, and the wasp *Nasonia vitripennis*, SCRMshaw successfully predicted enhancers in a cross-species fashion with an approximately 75% prediction success rate, based on reporter gene assays in xenotransgenic *D. melanogaster* (chosen as a pragmatic transgene host species) and comparison to already-identified enhancers in the other species (17–20). These results suggest that there are significant, albeit hidden, homologies governing the sequence characteristics of insect enhancers, at least for those involved in a substantial number of gene regulatory networks, and motivated us to apply SCRMshaw to a large and diverse set of sequenced insect genomes.

A cross-species SCRMshaw pipeline

To facilitate application of SCRMshaw to large numbers of newly-sequenced genomes, we developed a detailed workflow to ensure proper formatting of input genomes, rapid prediction of tissue-specific enhancer sequences, evaluation of results, and annotation of loci with information

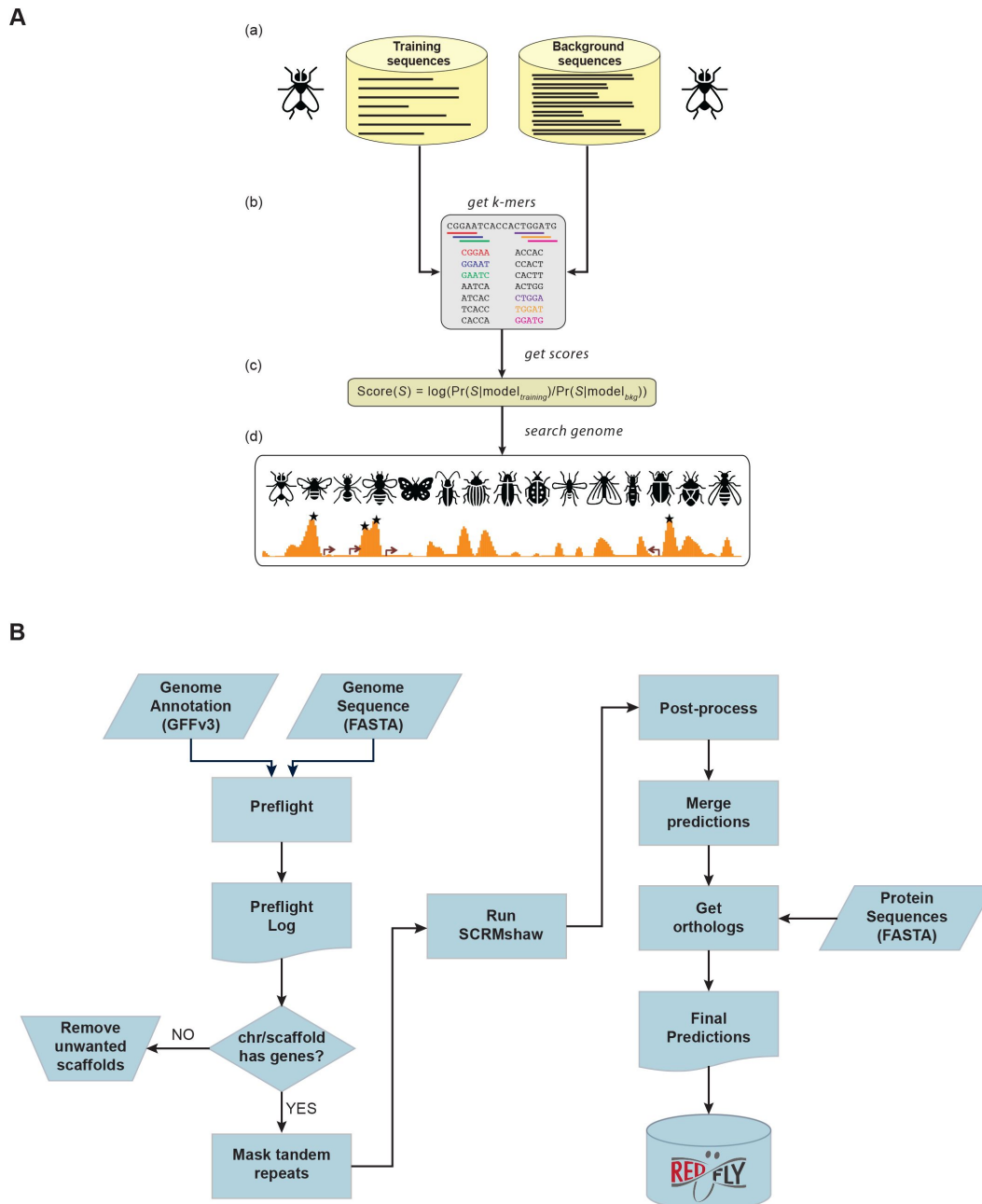


Figure 1

The SCRMshaw method and analysis pipeline.

(A) Supervised motif-blind CRM discovery (SCRMshaw). (a) SCRMshaw uses a training set of known *D. melanogaster* enhancers (“training sequences”), drawn from REDfly, that are defined by common functional characterization, and a 10-fold larger background set of similarly sized common functional characterization, non-enhancer sequences (“background sequences”). (b) The short DNA subsequence (*kmer*) count distributions of these sequences are then used to train a statistical model. The trained model (c) is used to score overlapping windows in the “target genome.” (d) High-scoring regions are predicted to be functional regulatory sequences (asterisks). Figure adapted from [73](#). (B) The workflow used for the regulatory genome annotation described in this paper. The left side shows pre-processing steps, the right side, post-processing. Input to SCRMshaw consists of the genome sequence and gene annotation. A protein sequence annotation is supplied later for the orthology mapping step. Final results are made available as part of the REDfly regulatory annotation knowledgebase.

drawn from the respective orthologous regions in the richly-investigated *D. melanogaster* genome (Fig. 1B; see Methods). The workflow consists of four major steps:

Preflight

SCRMshaw requires two input files for each genome: a FASTA-formatted file of the genome sequence itself, and a GFFv3-formatted file of the genome annotation. The genome file is masked for tandem repeats using Tandem Repeat Finder (22). *Preflight* validates the formats of these files and produces a comprehensive log file that highlights any issues along with basic information such as the number of chromosomes/scaffolds and their sizes, data types present in the annotation (e.g., ‘gene’, ‘exon’, ‘ncRNA’, etc.), and average intergenic distances. *Preflight* also provides a sample output of the SCRMshaw-generated ‘gene’ and ‘exon’ files. This feature allows users to identify any discrepancies or errors stemming from the input files and to reformat these files as needed before running SCRMshaw. Any minor scaffolds that are not annotated as containing genes are discarded.

SCRMshaw

SCRMshaw is run as previously described (15), using the “HD” variant (21).

Post-processing

The raw SCRMshaw output is post-processed to determine the final set of predicted enhancers. The original post-processing procedure described in (21) had a tendency to predict enhancers skewed toward long lengths (median ~ 1100 bp). We have revised that method here (see Methods) to yield predictions of more compact size (median 750 bp), which is more in keeping with empirically characterized enhancer lengths (23).

Orthology mapping

As a final step, we map all putative target genes in the SCRMshaw output to their *D. melanogaster* orthologs (if an ortholog exists) using a custom pipeline incorporating the *Orthologer* software from the Zdobnov lab (24) (see Methods). We use *D. melanogaster* as it has by far the most comprehensive gene annotation of the insects and thus provides the most detailed functional information for each gene. Mapping the genes from each species to a common ortholog allows us to assess whether we have obtained predicted enhancers in orthologous loci within the various species on which we have run SCRMshaw. The orthology mapping step requires that a set of predicted proteins is present as part of the existing genome annotation.

Annotation of 33 insect genomes

We ran our annotation pipeline on an initial set of 33 genomes (additional genome annotation is ongoing). These initial genomes were chosen based on availability and to sample broadly among the holometabolous insect orders and the Hemiptera (Fig 2A; Table 1). For each genome, we ran SCRMshaw using a collection of 48 training sets (Supplemental Table S1) and all three SCRMshaw scoring methods (“IMM,” “hexMCD,” “PAC-rc”; (13, 14)). For fifteen species where a protein annotation was available, we assigned *D. melanogaster* orthologs to each predicted locus, with an average of 54% (range 38-82%) of genes in a given species having a *D. melanogaster* ortholog (Fig. 2B).

Collectively, we predicted a total of 2,873,192 enhancers in these 33 species, with each species having on average approximately 87,000 predictions (Supplemental Table S2) (Fig. 2C). (As some enhancers are predicted by multiple scoring methods, or from more than one training set, the number of unique sequences is lower at 1,164,354; see gray bars in Fig. 2C.) The median length of the predicted enhancers across all species was 750 bp (mean 695, range 490-32500 bp). However, we noted the presence of a small number—2642, < 0.1% of the total—of unusually large regions (> 2000 bp), the bulk of which were confined to just a few genomes (Supplemental Fig. S1).

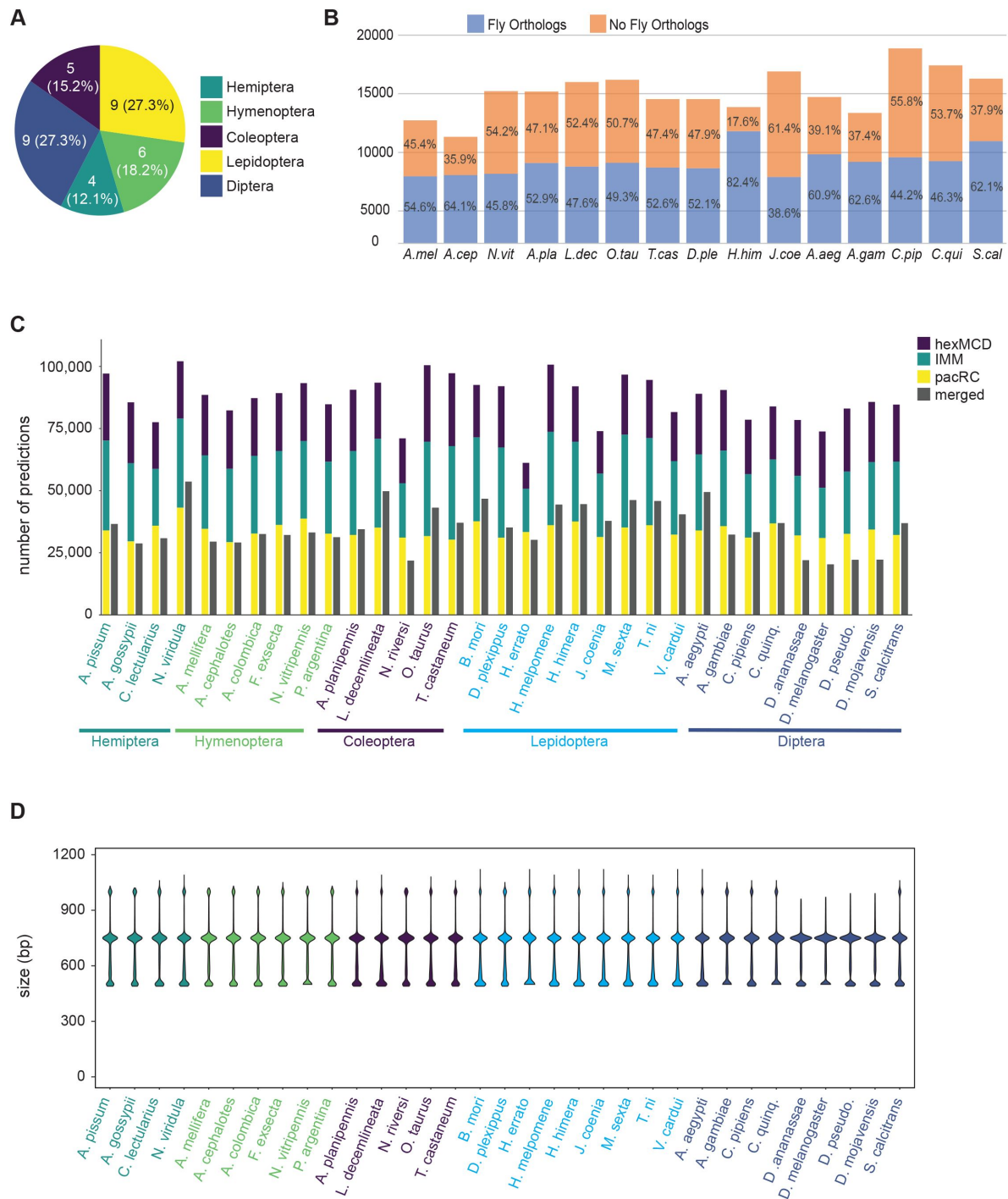


Figure 2

Annotation of 33 insect genomes.

(A) Genomes from five insect orders were annotated in this study (more are ongoing). (B) Percentage of genes with *Drosophila* orthologs as mapped via our orthology pipeline (see Methods), for the 15 mapped species. For complete species names, see **Table 1**. (C) Total SCRMshaw predictions for each species. For each species, the lefthand column shows cumulative results for each SCRMshaw sub-method summed over each of the 48 training sets. The righthand column shows the number of unique predictions after merging overlapping predictions from both sub-methods and training sets. Species are displayed alphabetically by taxonomic order. (D) Size distribution of SCRMshaw predictions, prior to merging overlapping predictions but after removing outlier predictions > 2 kb in length. Species are ordered identically to panel C.

Scientific name	common name	Order	Assembly version/ Annotation version	link/URL for assembly information
<i>Acyrtosiphon pisum</i>	pea aphid	Hemiptera	racon and v3_wdel	Courtesy Jennifer Brisson (University of Rochester)
<i>Aedes aegypti</i>	yellow fever mosquito	Diptera	L5.2	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7159/
<i>Agrilus planipennis</i>	emerald ash borer	Coleoptera	Apla_2.0	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/224129/
<i>Anopheles gambiae</i>	African malaria mosquito	Diptera	P4.9	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7165/
<i>Aphis gossypii</i>	cotton aphid/melon aphid	Hemiptera	ASM401081 v2	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/80765/
<i>Apis mellifera</i>	honey bee	Hymenoptera	HAv3.1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7460/
<i>Atta cephalotes</i>	leafcutter ant	Hymenoptera	A.ceph_1.0	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/12957/
<i>Atta colombica</i>	leafcutter ant	Hymenoptera	Acol1.0	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/520822/
<i>Bombyx mori</i>	silkworm	Lepidoptera	ASM15162v1	http://ensembl.lepbase.org/Bombyx_mori_asm15162v1/Info/Index
<i>Cimex lectularius</i>	bed bug	Hemiptera	Clec_2.1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/79782/
<i>Culex pipiens pallens</i>	Northern house mosquito	Diptera	TS_Cpip_V1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/42434/
<i>Culex quinquefasciatus</i>	Southern house mosquito	Diptera	VIPSU_Cqui_1.0_pri_parnal	https://academic.oup.com/gbe/article/13/3/evab005/6121099
<i>Danaus plexippus</i>	monarch butterfly	Lepidoptera	v3	http://ensembl.lepbase.org/Danaus_plexippus_v3/Info/Index
<i>Drosophila ananassae</i>	fruit fly	Diptera	caf1	http://ftp.flybase.net/genomes/Drosophila_ananassae/
<i>Drosophila melanogaster</i>	fruit fly	Diptera	r6 1.8	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7227/
<i>Drosophila mojavensis</i>	fruit fly	Diptera	caf1	https://www.ncbi.nlm.nih.gov/assembly/GCF_000005175.2/
<i>Drosophila pseudoobscura</i>	fruit fly	Diptera	r3	http://ftp.flybase.net/genomes/Drosophila_pseudoobscura/dpse_r3.03_FB2015_01/

Table 1

Species used in this study

<i>Formica exsecta</i>	wood ant	Hymenoptera	ASM365146 v1	https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_003651465.1/
<i>Heliconius erato</i>	red postman butterfly	Lepidoptera	v1	http://ensembl.lepbase.org/Heliconius_erato_lativitta_v1/Info/Index
<i>Heliconius melpomene</i>	postman butterfly	Lepidoptera	Hmel2	http://ensembl.lepbase.org/Heliconius_melpomene_melpomene_hmel2/Info/Index
<i>Heliconius himera</i>	false postman butterfly	Lepidoptera	Hed.V1	Courtesy Robert Reed (Cornell University)
<i>Junonia coenia</i>	common buckeye butterfly	Lepidoptera	JC v1.0	Courtesy Robert Reed (Cornell University)
<i>Leptinotarsa decemlineata</i>	Colorado potato beetle	Coleoptera	Ldec_2.0	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7539/
<i>Manduca sexta</i>	tobacco hornworm	Lepidoptera	v1.0	http://ensembl.lepbase.org/Manduca_sexta_msex1/Info/Index
<i>Nezara viridula</i>	southern green stink bug	Hemiptera		https://www.ncbi.nlm.nih.gov/datasets/taxonomy/85310/
<i>Nasonia vitripennis</i>	jewel wasp	Hymenoptera	Psr_1.1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7425/
<i>Onthophagus taurus</i>	dung beetle	Coleoptera	Otau_2.0	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/166361/
<i>Pseudoatta argentina</i>	leafcutter ant	Hymenoptera	ASM176075 2v1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/621737/
<i>Stomoxys calcitrans</i>	stable fly	Diptera	Stomoxys_calcitrans-1.0.1	https://bmcbiol.biomedcentral.com/articles/10.1186/s12915-021-00975-9?sap-outbound-id=DB0B95C63ABC56834FFE23B3C176FAE498F3D66E&utm_source=hybris-campaign&utm_medium=email&utm_campaign=000_MKQ5584_0000012067_BMCF_AWA_MK01_GL_12915_eToc_Apr21_CTRL&utm_content=EN_internal_24671_20210419&mkt-key=42010A0557EB1EDA9CFC80777B53EE6F
<i>Tribolium castaneum</i>	red flour beetle	Coleoptera	r5.2	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7070/
<i>Trichoplusia ni</i>	cabbage looper	Lepidoptera	tn1	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/7111/
<i>Vanessa cardui</i>	painted lady butterfly	Lepidoptera	Vcar_v1	Courtesy Robert Reed (Cornell University)
<i>Nebria riversi</i>	ground beetle	Coleoptera	v1	https://onlinelibrary.wiley.com/doi/abs/10.1111/1755-0998.13409

Table 1 (continued)

We therefore discarded any predictions with length greater than 1.5 times the interquartile range of the complete prediction set and re-evaluated the size distribution. This resulted in a median size of 740 bp with a mean of 676 bp and a range of 490-1120 bp, indicating that the overall impact of these outlier sequences is minimal (**Fig. 2D**). Inspection of the excessively large elements revealed that they result from SCRMshaw predictions that lie immediately adjacent to each other (without gaps) and have similar SCRMshaw scores, which thus become merged into one broad predicted element. Preliminary analysis suggests that these regions result from insufficient masking of tandem repeat regions and/or genome assembly errors, although other causes, such as extremely enhancer-dense “superenhancer” regions (reviewed by 25), cannot entirely be ruled out.

Many loci contain multiple enhancer predictions

We have noted in the past that SCRMshaw often predicts multiple enhancers in a single locus (e.g. 26). This is consistent with the concept of “shadow enhancers,” sets of multiple redundant or semi-redundant enhancers regulating the same gene (reviewed by 27). On the other hand, if SCRMshaw is predicting enhancers with low specificity (i.e., largely at random), it is instead possible that larger loci may just accumulate a high number of predictions simply due to their greater length.

To distinguish between these possibilities, we conducted simulations on three representative genomes (*D. melanogaster*, *C. pipiens*, and *A. aegypti*), each with a different average intergenic region size chosen to represent small, medium and large genomes respectively (average sizes 610 bp, 5,274.5 bp, and 14,641 bp; see Methods). We randomized the location of SCRMshaw predictions across the non-coding component of each genome and compared the randomized results to our SCRMshaw output. We found that, for a subset of loci, the number of real SCRMshaw predictions per locus was consistently higher than the number obtained at random (Supplemental Table S3). For example, using the *mapping1.visceral_mesoderm* training set, 4.6% (15/328) of *D. melanogaster* loci containing one or more SCRMshaw predictions had a significantly ($P < 0.001$) larger number of predictions/locus than expected, 1.5% (10/658) of *C. pipiens* loci had more predictions than expected, and 1% (3/299) of *A. aegypti* loci had excess predicted enhancers (Supplemental Table S3). Similarly, for the *mapping2.ectoderm* training set, 7.6%, 1.5%, and 11% of loci in the three species, respectively, had a significantly ($P < 0.001$) greater than expected number of predictions per locus (Supplemental Table S3). Averaged over all training sets, 3-4% of loci with predictions had significantly more than expected by chance (mean values: 3.0% for *D. melanogaster*, 3.0% for *C. pipiens*, and 3.7% for *A. aegypti*; maximum values: 10.2% for *D. melanogaster*, 9.3% for *C. pipiens*, and 11.6% for *A. aegypti*). Only very few training sets did not have any loci with significantly more enhancers than expected (1 set in *A. aegypti*, 1 in *C. pipiens* and 8 in *D. melanogaster*; Supplemental Table S3).

To further ensure that these results were not influenced by locus size, we binned the loci by length and assessed the numbers of real versus simulated predictions/locus for each bin. The binned results were similar to the results using all loci, i.e., the number of SCRMshaw predictions/locus (**Fig. 3**, blue boxplots) was consistently higher than the number obtained via randomization (**Fig. 3**, pink boxplots). Results were also similar when comparing predictions only at intergenic versus only at intronic positions (Supplemental Table S3). These results suggest that SCRMshaw is not predicting enhancers randomly throughout the genome, but rather is identifying multiple related “shadow” enhancers in a subset of loci. Consistent with this interpretation, functional tests of the full set of predicted enhancers in several *D. melanogaster* loci has confirmed that many of the predictions act as functionally similar enhancers (T. Williams, personal communication).

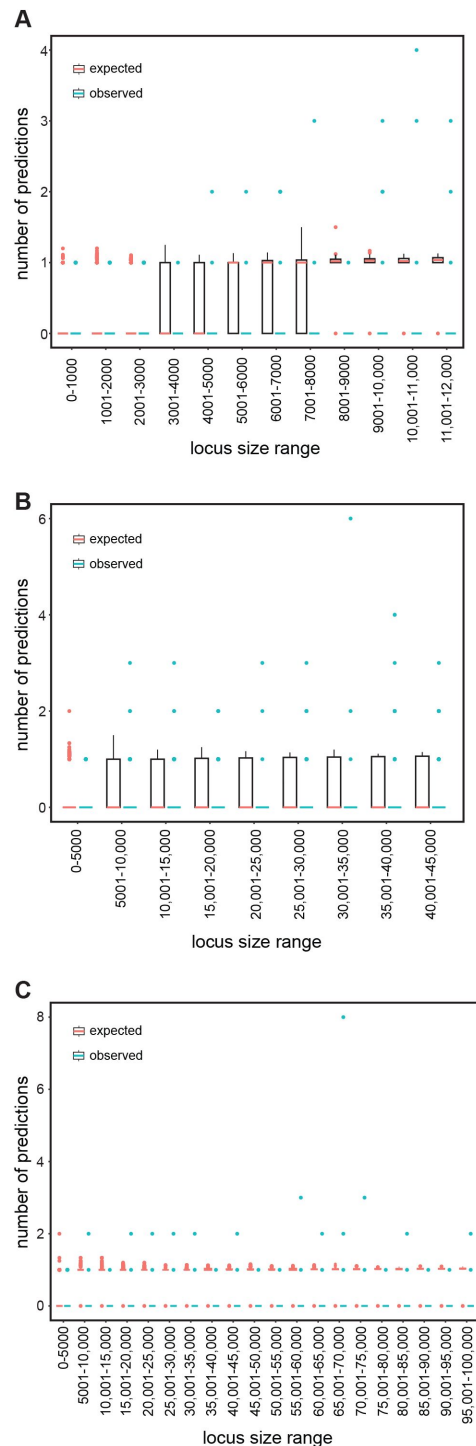


Figure 3

SCRMshaw makes multiple predictions per locus.

The number of SCRMshaw predictions per locus (y-axis) are shown as boxplots for loci falling within the given size ranges (x-axis). Black boxes cover the 25th-75th percentiles, bars indicate median values and dots indicate values exceeding 1.5 times the interquartile range (boxes are not visible for all bins due to very low degrees of variation). Values in pink represent expected values drawn from randomization, while values in blue represent observed values from SCRMshaw. All values are from results with the training set “mapping1.visceral_mesoderm”; results from other training sets were similar (see Supplemental Table S3). Shown are results from the genomes of (A) *D. melanogaster*, (B) *C. pipiens*, and (C) *A. aegypti* representing small, medium, and large genomes, respectively.

SCRMshaw predicts enhancers in orthologous loci across species

A major premise underlying cross-species applications of SCRMshaw is that conserved regulatory strategies should allow for a model trained on enhancers in one species to predict for similar enhancers in another species. We reasoned that at least some of the time, this should lead to identification of enhancers in orthologous loci, as orthologous genes are frequently involved in similar biological pathways and developmental regulatory networks (30). Indeed, we previously showed that SCRMshaw was able to predict enhancers in several orthologous loci, for instance those for the *single-minded* locus in *D. melanogaster*, *A. gambiae*, and *A. mellifera*, and the *wingless* locus in *D. melanogaster*, *A. mellifera*, and *T. castaneum* (17). To test whether this is generally true, we used the fifteen species with mapped *D. melanogaster* orthologs (plus *D. melanogaster* itself), and all 48 of our training sets, to compare the number of SCRMshaw-based versus random predictions obtained in common (orthologous) loci.

We observed a substantial reduction in the number of orthologous loci with predictions in common as we moved from considering a set of ten to the full set of all sixteen species (average number of common loci over all 48 training sets: 70.7 (10 species) > 39.4 > 19.8 > 9.14 > 4.12 > 1.6 > 0.6 (16 species) out of an average number of 640 predictions with mapped orthologs) (Fig. 4A, Supplemental Table S4a). This rapid decline is likely due to a variety of factors, including differences in taxonomic order (e.g., Diptera vs. Hymenoptera), the quality and degree of ortholog data for each species, and the quality of annotation for each species, all of which likely lead to an underestimation of the true numbers of common loci in our data (see Discussion). We simulated SCRMshaw predictions for all sixteen species by randomizing the SCRMshaw results and compared the number of common loci between the simulated and real data for combinations of ten to sixteen species. The number of common loci among the real SCRMshaw predictions was consistently significantly higher than the number of common loci observed in the simulated data ($P < 0.05$; Fig. 4B and Supplemental Table S4). In particular, when evaluating between 10 and 12 species (data for 14–16 species are not reliable due to the very small number of observed loci in common) only a single training set, *adult_PNS*, did not have a significant overrepresentation of common loci (Supplemental Table S4c). We also examined the fold enrichment, i.e., the extent to which the true number was in excess of the simulated number, and found that on average there were greater than 2.4x more predictions in common loci than expected, when considering groups of 10–12 species; almost all training sets had a fold enrichment >1.5x (Fig. 4C, Supplemental Table S4d). These results strongly suggest that SCRMshaw is successfully finding sets of conserved (or convergent) enhancers, as it is predicting sequences in orthologous loci in a training-set specific manner across multiple species significantly more often than can be accounted for by chance.

SCRMshaw predictions correlate with regions of accessible chromatin

Active enhancers are frequently found in regions of accessible chromatin, as assayed by methods such as DNase-seq, FAIRE-seq (Formaldehyde-Assisted Isolation of Regulatory Elements with sequencing), and ATAC-seq (Assay for Transposase-Accessible Chromatin by sequencing) (28–30). We confirmed previously that FAIRE-predicted and SCRMshaw-predicted enhancers in *T. castaneum* have a high (>79%) degree of overlap (18). To determine whether a similar correlation exists for other species, we compared our SCRMshaw predictions with available FAIRE-seq and ATAC-seq data for seven species: *D. melanogaster*, *T. castaneum*, *A. gambiae*, *Danaus plexippus*, *Vanessa cardui*, *Junonia coenia*, and *Heliconius himera*. These comparisons are imperfect, as the tissues used to obtain the chromatin data do not precisely correspond to the training sequences used for SCRMshaw. Nevertheless, in the majority of cases where we were able to establish a rough match between the tissues, we observed significant overlap between the two

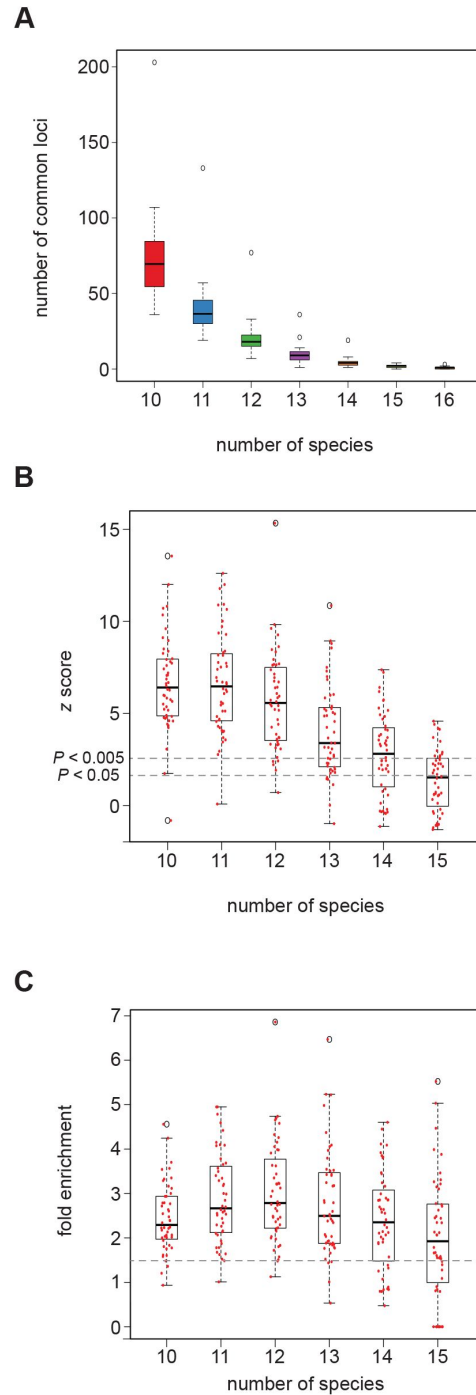


Figure 4

SCRMshaw predicts CRMs in orthologous loci across species.

(A) The number of loci in common that contain at least one SCRMshaw prediction, for 10 or more species. (B) z- scores demonstrating that the number of loci in common with one or more SCRMshaw predictions is significantly higher than expectation, based on 360 randomizations. The small number of common predictions for 14-16 species make these statistics unreliable. Dotted lines indicate z-score values representing significance at the (unadjusted) $P < 0.005$ and $P < 0.05$ levels. (C) Fold enrichment values illustrating the excess of loci in common with one or more SCRMshaw predictions compared to expectation. Dotted line shows 1.5x enrichment.

methods of enhancer detection. For example, in *D. melanogaster*, 40% of SCRMshaw predictions using the *blastoderm.mapping1* training set overlapped ATAC-seq peaks obtained from blastoderm embryos ($P < 4.15 \times 10^{-112}$, fold enrichment 2.98) (31) (Table 2 row 2). Similarly, 35% and 41% of SCRMshaw predictions from the *mapping2.wing* and *haltere_disc* sets overlapped FAIRE data that included wing and haltere cells (32) ($P < 4.6 \times 10^{-117}$ and 1.5×10^{-102} , fold enrichment of 3.29 and 2.98) (Table 2, rows 1, 3). In the mosquito *A. gambiae*, we compared SCRMshaw predictions from the *embryonic_midgut* and *mapping1.salivary* training sets to ATAC-seq data from adult midgut and salivary tissues, observing overlaps of 40% and 37% respectively ($P < 1.5 \times 10^{-102}$ and 2.15×10^{-93} , fold enrichment of 3.45 and 3.22;) (Table 2, rows 9, 10). When comparing our SCRMshaw predictions from the *mapping2.wing* set to ATAC-seq data for larval wing tissue in the butterflies *D. plexippus* and *H. himera*, we observed overlaps of 60% and 68% ($P < 5.59 \times 10^{-19}$ and $P < 9.92 \times 10^{-25}$, fold enrichment of 1.99, 1.66;) (Table 2, rows 11, 12) (33). The butterflies *J. coenia* and *V. cardui* are exceptions; intriguingly, they show a depletion in SCRMshaw predictions compared to expectation (Table 2, rows 13, 14). Whether this is due to a mismatch in the data used for comparison, the state of the genome assemblies for these two draft genomes, or some other failure of SCRMshaw to perform strongly on these species remains to be determined. Overall, the highly significant overlap we observe between SCRMshaw predictions and open chromatin regions in most species and tissues provides further strong evidence that SCRMshaw is effectively predicting enhancers across a broad range of genomes.

Reporter gene analysis demonstrates that SCRMshaw predictions are functional enhancers

As a concrete test of our ability to use SCRMshaw to predict functional enhancers, we assayed a subset of SCRMshaw predictions by reporter gene analysis in transgenic *D. melanogaster*. Our previous studies have shown a high success rate for such assays, ranging from 65% through >80%, depending on the species and training sets used (13, 14, 17, 19, 20).

We focused our *in vivo* validation experiments on SCRMshaw predictions made using the *mapping2.wing*, *haltere_disc*, and *disc.mapping2* training sets and a set of four species we had previously shown to be amenable to SCRMshaw prediction: *D. melanogaster*, *A. aegypti*, *T. castaneum*, and *A. mellifera*. The imaginal discs are well-established tissues for investigating gene regulation in *D. melanogaster*, and the chosen training sets all gave significant results in the open-chromatin comparisons discussed above. We selected six sets of putative enhancers for testing (Table 3, Table 4). For the first three sets, we selected sequences where we had predictions in each of the orthologous loci for *D. melanogaster*, *T. castaneum*, and *A. mellifera* (*A. aegypti* was not considered for these sets), and where the *D. melanogaster* prediction mapped near a gene expressed in the wing imaginal discs (Fig. 5). These predictions were conducted using SCRMshaw's IMM scoring method only, with post-processing performed using the original method described in (21), and the Amel_4.5 version of the *A. mellifera* genome. For the second three sets, we chose sequences where we had predictions in orthologous loci for at least three of the four species and where the *D. melanogaster* sequence had previously been tested in a reporter gene assay and was known to be active in the relevant imaginal discs (Fig. 5). Imposing this latter criterion allowed us to leverage existing knowledge and reduce the necessary amount of *in vivo* testing for each set of predictions, enabling us to test a larger set of sequences overall. For this second set of three, we used all three SCRMshaw scoring methods with a revised post-processing algorithm (see Methods), and the Amel_Hav3.1 *A. mellifera* genome. For all six sets, predictions were chosen for testing based on high SCRMshaw scores and overlap with open-chromatin data (where available). We also considered the position of each prediction within the locus (i.e., first intron, downstream intergenic region, etc.), favoring sequences where position was maintained among the orthologs. Selected sequences were cloned into a cross-species compatible reporter vector (Deem et al., submitted) (18), and reporter gene activity was visualized either directly or by using the lineage tracing system G-TRACE (34). In the latter, enhancer activity is visualized

Species	Training Set	Profiled Tissue	Overlap (real)	Overlap (random)	s.d. ^a	Z-score	FE ^b
<i>D. melanogaster</i>	haltere_disc	wing, leg, and haltere third instar discs and pharate appendages; eye-antennal disc; third instar CNS	41.82%	14.02%	10.65	21.53	2.98
<i>D. melanogaster</i>	blastoderm.mapping1	blastoderm	40.99%	13.86%	12.35	22.56	2.96
<i>D. melanogaster</i>	mapping2.wing	wing, leg, and haltere third instar discs and pharate appendages; eye-antennal disc; third instar CNS	35.14%	10.69%	10.19	23.01	3.29
<i>T. castaneum</i>	mapping2.wing	embryo, larval thoracic epidermis, larval brain	85.58%	26.18%	10.98	28.88	3.27
<i>T. castaneum</i>	mapping2.ectoderm	embryo, larval thoracic epidermis, larval brain	81.68%	25.48%	12.48	26.29	3.21
<i>T. castaneum</i>	mapping1.ventral_ectoderm	embryo, larval thoracic epidermis, larval brain	81.17%	24.76%	12.56	33.87	3.28
<i>T. castaneum</i>	mapping1.dorsal_ectoderm	embryo, larval thoracic epidermis, larval brain	71.15%	24.69%	12.64	24.07	2.88
<i>T. castaneum</i>	mapping1.ectoderm	embryo, larval thoracic epidermis, larval brain	69.53%	24.66%	11.92	22.35	2.82
<i>A. gambiae</i>	embryonic_midgut	adult midgut, salivary_gland	40.20%	11.66%	7.90	21.56	3.45
<i>A. gambiae</i>	mapping1.salivary_gland	adult midgut, salivary_gland	36.76%	11.43%	8.52	20.55	3.22
<i>D. plexippus</i>	mapping2.wing	larval forewing, hindwing, head	60.43%	30.35%	6.30	8.93	1.99
<i>H. himera</i>	mapping2.wing	larval forewing, hindwing, head	68.34%	41.10%	8.97	10.27	1.66
<i>J. coenia</i>	mapping2.wing	larval forewing, hindwing, head	3.06%	34.91%	8.52	-12.22	0.09
<i>V. cardui</i>	mapping2.wing	larval forewing, hindwing, head	15.46%	36.01%	8.01	-7.13	0.43

^as.d., standard deviation
^bFE, fold enrichment

Table 2

Overlap of SCRMshaw predictions with FAIRE-seq and ATAC-seq peaks

through two reporters; the first reporter visualizes the direct enhancer activity while expression of the second reporter is induced and maintained in all cells that descend from a cell in which the enhancer is initially active, even if activity subsequently shuts off.

ex

The *expanded* (*ex*) gene plays a crucial role in tissue growth control, including wings (35–37). There was a single prediction in the *D. melanogaster* *ex* locus, *Dm_ex_20p0*, which falls within an open chromatin region of the third intron (Supplemental Fig. S2), and which comprises an untested subsequence of a longer sequence that acts as an imaginal disc enhancer (Fig. 5B; (37)). This sequence drove reporter activity in the wing, leg, and antennal discs (Fig. 6A). The *T. castaneum* genome also had only one prediction for the *ex* locus, *Tc_ex_9p0*, which overlaps well with a FAIRE-seq peak (Supplemental Fig. S2) within the second intron. Although no imaginal disc activity was observed (Fig. 6B), *Tc_ex_9p0*-driven reporter gene expression was observed during late pupal stages in the legs (Supplemental Fig. S4G). The *A. mellifera* genome (v4.5) had one prediction, *Am_ex_20p3*, within the fourth intron of the *ex* locus (Supplemental Fig. S2; however, note that subsequent prediction using the updated Amel_Hav3.1 genome and revised SCRMshaw post-processing yielded additional predictions in this locus). *Am_ex_20p3* drove active but variable expression in both the pouch and notum regions of the wing imaginal disc (Fig. 6C). Although often significantly limited to a small number of cells, the pouch expression of *Am_ex_20p3* was similar in pattern to that seen with *Dm_ex_20p0* (cf. Fig. 6A).

Weak and inconsistent expression in the pouch region of the haltere discs was also observed. In addition, *Am_ex_20p3* drove expression in the leg discs (Fig. 6C).

klu

Klumpfuss (*klu*) encodes a zinc finger protein important for proper tissue specification and differentiation (38). The *D. melanogaster* genome had three predictions for the *klu* locus (Supplemental Fig. S2). We selected *Dm_klu_16p1*, which overlaps a region of open chromatin within the second intron present in most imaginal discs (most distinct in the T3 leg disc; Supplemental Fig. S2). *Dm_klu_16p1* drove active expression in the notum regions of the wing and haltere discs and in the leg discs (Fig. 6D). *Tc_klu_8p6* was the only prediction for the *T. castaneum* *klu* locus. It falls within the second intron, similar to *D. melanogaster*'s *klu* prediction *Dm_klu_161p1* (Supplemental Fig. S2), but lacked enhancer activity (Fig. 6E). *A. mellifera* had only one prediction for the *klu* locus (*Am_klu_20p2*) (Supplemental Fig. S2). We included this prediction for validation even though its location in the third intron does not match with that in the other two species. *Am_klu_20p2* drove weak expression in the most proximal part of the leg discs (Fig. 6F, arrows), but no activity was observed in the other imaginal discs. (Note that additional predictions were subsequently produced with our revised post-processing algorithm and the updated Amel_Hav3.1 genome, for both the *T. castaneum* and *A. mellifera* *klu* loci).

ush

Our final choice from our first set for *in vivo* validation was *u-shaped* (*ush*), which encodes a transcription factor with described imaginal disc expression (39–41). The *D. melanogaster* genome had two predictions for the *ush* locus (three when using the updated post-processing algorithm) (Supplemental Fig. S2). We tested *Dm_ush_16p4*, but did not observe any larval disc activity (Fig. 6G). *Tc_ush_6p8* and *Am_ush_20p8* were the only predictions for *T. castaneum* and *A. mellifera*, respectively, both located in the second intron (again, additional predictions are found using updated methods and genome versions). *Tc_ush_6p8* had expression in the wing disc as well as weak activity in the peripodial membrane of the eye-antennal disc (Fig. 6H, Supplemental Fig. S4K). *Am_ush_20p8* displayed active expression in the peripodial membrane surrounding the eye disc, as well as in a single cell, or small subset of cells, located at the center of each leg disc (Fig. 6I, arrows).

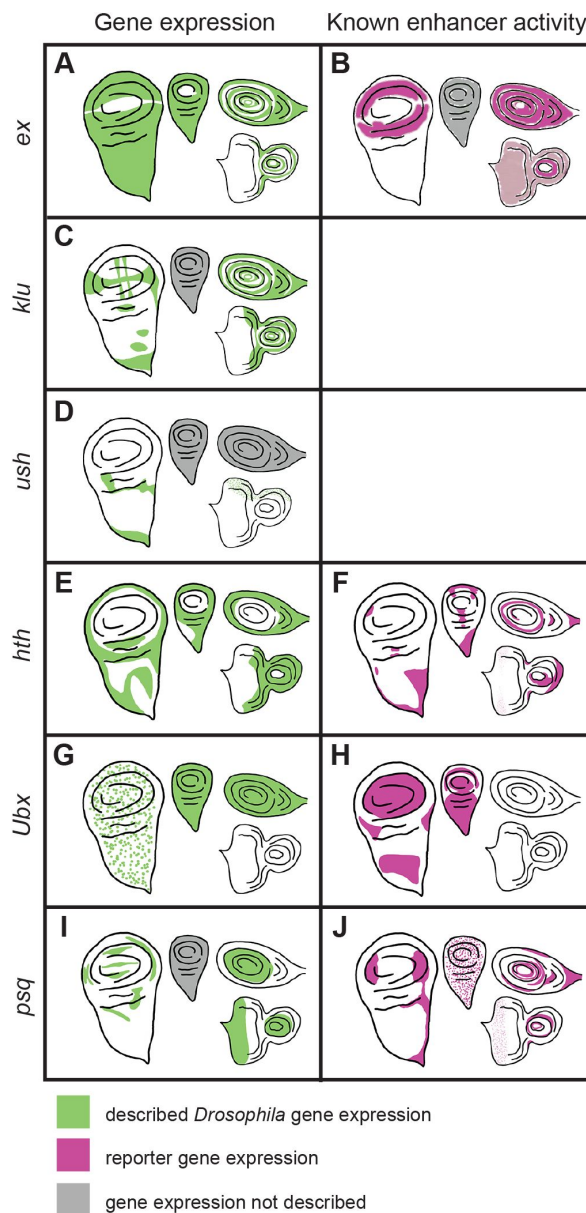


Figure 5

Previously described gene expression and enhancer activity for select *D. melanogaster* sequences predicted by SCRMshaw.

The lefthand column shows native *D. melanogaster* gene expression in imaginal discs (green), while the righthand column shows described enhancer activity (magenta). Gray shading indicates that expression has not been described. Moving clockwise from the left side of each panel are the wing, haltere, leg, and eye- antennal discs. The enhancers whose activities are described in the table are: (B) *ex_BCDE* (37), (F) *hth_GMR46D04* (42), (H) *Ubx_GMR39A02* (42), (J) *psq_GMR41E12* (42).

<i>D. melanogaster</i>			<i>T. castaneum</i>		<i>A. mellifera</i>	<i>A. aegypti</i>
name	symbol	FlyBaseID	RefSeq	iBeetleBase	RefSeq	VectorBase
<i>expanded</i>	<i>ex</i>	FBgn0004583	<i>gene-LOC657053</i>	TC012545	<i>gene-LOC551519</i>	AAEL001437
<i>u-shaped</i>	<i>ush</i>	FBgn0003963	<i>gene-LOC659918</i>	TC013689	<i>gene-LOC100577801</i>	AAEL020615
<i>klumpfuss</i>	<i>klu</i>	FBgn0013469	<i>gene-LOC103312803</i>	TC002783	<i>gene-LOC100577692</i>	AAEL013544
<i>homothorax</i>	<i>hth</i>	FBgn0001235	<i>gene-Hth</i>	TC008629	<i>gene-LOC552079</i>	AAEL011643
<i>pipsqueak</i>	<i>psq</i>	FBgn0263102	<i>gene-LOC660343</i>	TC003349	<i>gene-psq</i>	AAEL021255
<i>Ultrabithorax</i>	<i>Ubx</i>	FBgn0003944	<i>gene-Ubx</i>	TC000903	<i>gene-ubx</i>	AAEL014032

Table 3

Gene loci chosen for *in vivo* validation

Coordinates		Max. score	Training set(s)	Method
Set 1				
<i>T. castaneum</i>				
Tc_ex_9p0	NC_007424.3:11221530..11222170	9.04	mapping2.wing	imm
Tc_ush_6p8	NC_007420.3:5968840..5969370	6.84	haltere_disc	imm
Tc_klu_8p6	NC_007418.3:10416700..10417300	8.59	mapping2.wing	imm
<i>D. melanogaster</i>				
Dm_ex_20p0	2L:442110..442810	20.04	mapping2.wing	imm
Dm_klu_16p1	3L:10991040..10991700	16.06	mapping2.wing	imm
Dm_ush_16p4	2L:531250..532060	16.40	mapping2.wing	imm
<i>A. mellifera</i>				
Am_ex_20p3	NC_007075.3:7545750..7546440	20.36	mapping2.wing	imm
Am_klu_20p2	NC_007070.3:720220..720870	20.25	mapping2.wing	imm
Am_ush_20p8	NC_007080.3:10609550..10610200	20.80	haltere_disc	imm
Set 2				
<i>T. castaneum</i>				
Tc_hth_15p5	NC_007422.5:13408990..13409850	15.52	mapping2.wing	hexmcd,imm,pac
Tc_psq_19p7	NC_007418.3:1805000..1806250	19.71	haltere_disc, mapping2.wing	hexmcd,imm,pac
Tc_Ubx_19p9	NC_007417.3:8137250..8138000	19.32	haltere_disc	hexmcd
Tc_Ubx_17p4	NC_007417.3:8153250..8154030	19.95	mapping2.wing	hexmcd,imm
<i>A. aegypti</i>				
Aa_hth_35p9	1:149733960..149734710	35.96	mapping2.wing	imm
Aa_psq_21p5	2:228190750..228191640	21.50	disc.mapping2	hexmcd,imm
Aa_Ubx_26p0	1:309747490..309748390	26.08	mapping2.wing	imm
<i>A. mellifera</i>				
Am_psq_29p2	NC_007078.3:10712000..10712750	29.26	mapping2.wing	hexmcd
Am_Ubx_37p2	NC_007085.3:2921250..2922250	37.18	haltere_disc, mapping2.wing	hexmcd,imm
Am_Ubx_0p39	NC_007085.3:2967750..2968250	0.39	mapping2.wing	pac

Table 4

SCRMshaw predictions chosen for *in vivo* validation

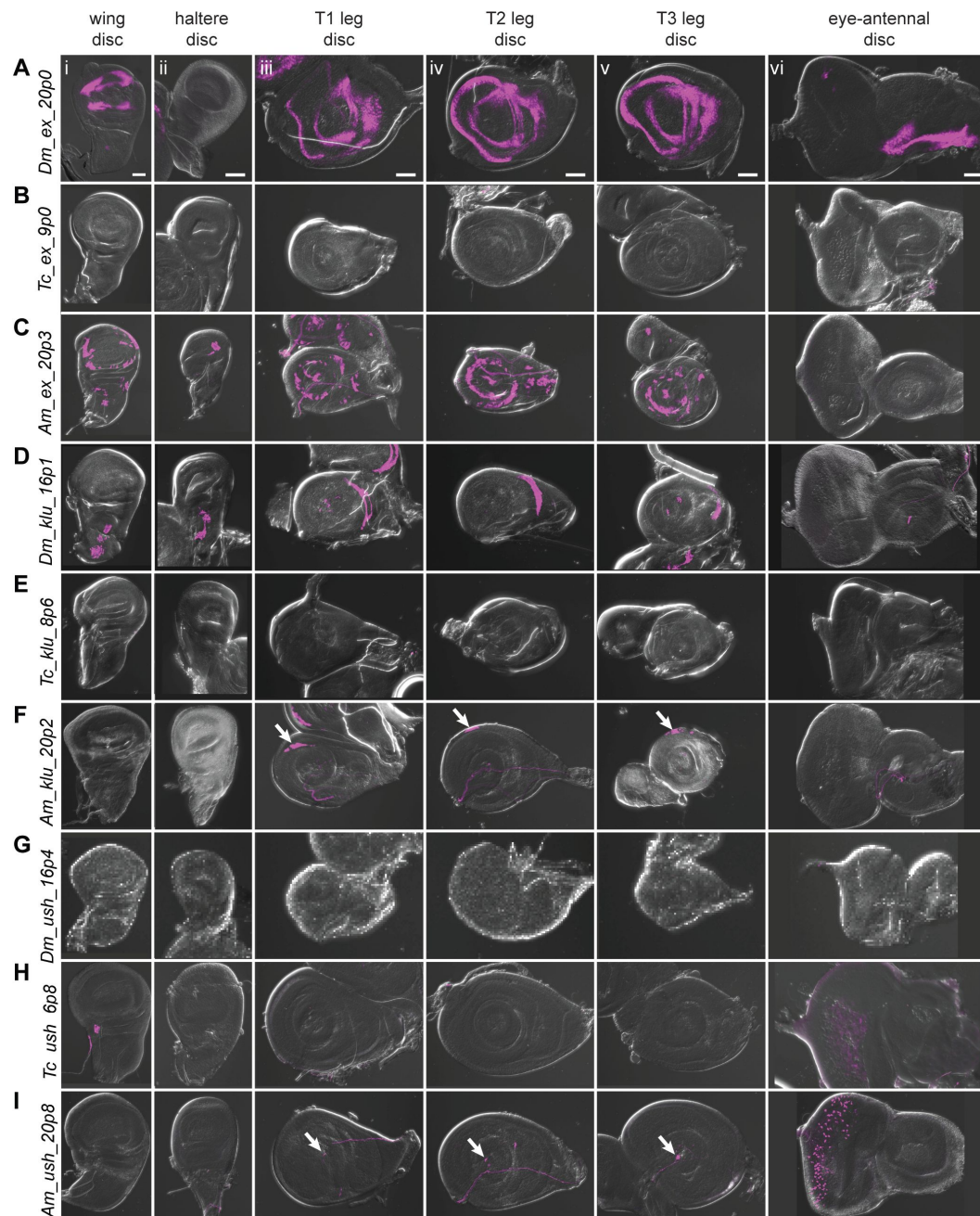


Figure 6

Reporter gene expression for tested *ex*, *klu*, and *ush* predicted enhancer sequences.

Each row shows expression for the indicated construct in (i) wing discs, (ii) haltere discs, (iii) T1 (prothoracic) leg discs, (iv) T2 (mesothoracic) leg discs, (v) T3 (metathoracic) leg discs, and (vi) eye-antennal discs (with eye portion to the left). Enhancer activities were visualized by UAS-tdTomato that was included in the reporter construct. Scale bar is 50 μ m for each column.

hth

SCRMshaw predicted ten enhancers in the locus of the *D. melanogaster* Hox cofactor *homothorax* (*hth*). One of the two top-scoring predictions, *Dm_hth_30p1*, located in the fifth intron, overlapped the known *hth_GMR46D04* enhancer, which has activity in all of the larval imaginal discs (Fig. 5F, Supplemental Fig. S3)(42). *A. aegypti* had a single prediction in the orthologous *hth* locus, *Aa_hth_35p9*, located in the first intron (Supplemental Fig. S3). This sequence failed to display activity in imaginal discs (Fig. 7A).

A. T. castaneum had 23 *hth*-locus predictions. We selected a high-scoring (albeit not the highest-scoring) sequence, *Tc_hth_15p5*, due to the similarity of its position to that of the *D. melanogaster* enhancer—in the fifth intron—and overlapping FAIRE peak (Supplemental Fig. S3)(18). The *Tc_hth_15p5* reporter showed activity in the presumptive notum region of the wing disc and in the proximal region of the leg discs, resembling both the endogenous *D. melanogaster hth* expression pattern and that of the *GMR46D04* enhancer (Fig. 7B, cf. Fig. 5F). No enhancer activity was observed in the haltere or eye-antennal discs (Fig. 7B). In the pupal stage, *Tc_hth_15p5* exhibited active expression along the thorax, corresponding to adult *hth* expression in *D. melanogaster* (Supplemental Fig. S4I, arrows) (43). No *hth* predictions were obtained for *A. mellifera*.

Ubx

The classic homeotic gene *Ultrabithorax* (*Ubx*) regulates tissue identity in the thoracic and abdominal segments (44). Of six SCRMshaw predictions in the *D. melanogaster Ubx* locus, we noted that one, *Dm_Ubx_36p1*, overlaps a cluster of known enhancers in the third intron centered on *Ubx_GMR39A02* and *Ubx_abx6.8* (Supplemental Fig. S3). *Ubx_GMR39A02* mirrors the haltere activity of *Ubx*, but also displays ectopic activity in the pouch and notum regions of the wing disc (Fig. 5H)(42). Similarly, *Ubx_abx6.8* also drives both native and ectopic expression in the imaginal discs (45).

There were eight predictions in the *Ubx* locus of the *A. aegypti* genome (Supplemental Fig. S3). We selected sequence *Aa_Ubx_26p0*, as it was both the highest scoring prediction and was in the third intron, similar to its putative *D. melanogaster* counterparts. Although direct reporter expression from *Aa_Ubx_26p0* was too weak to observe directly, use of the lineage-tracing G-TRACE system confirmed widespread activity in all leg discs (Fig. 7C).

We chose two *T. castaneum* sequences for testing, out of ten predictions: *Tc_Ubx_17p4*, which overlapped well with an accessible chromatin region in the third thoracic epidermal tissue and the central nervous system (18), and *Tc_Ubx_19p9*, which had the highest local prediction score but only overlapped with a chromatin region accessible predominantly during early embryogenesis (Supplemental Fig. S3). *Tc_Ubx_17p4* displayed activity in all three leg discs (Fig. 7D). The expression in T2 and T3 leg discs corresponds to endogenous *Ubx* expression, whereas T1 leg disc expression appears to be ectopic. *Tc_Ubx_19p9*, predicted using the “haltere” training set, did not drive clear expression in the haltere disc but did drive G-TRACE expression in the wing disc in the adepithelial adult myoblast cells, as well as direct reporter expression in the peripodial membrane of the eye disc. Very limited expression was also observed in the leg discs when using G-TRACE (Fig. 7E, Supplemental Fig. S4J).

The *A. mellifera* genome had eleven predictions at the *Ubx* locus (Supplemental Fig. S3). *Am_Ubx_37p2* was chosen based on a high SCRMshaw score, although it falls within the fourth, rather than the third, intron. *Am_Ubx_0p39*, on the other hand, was in the corresponding third intron location. *Am_Ubx_0p39* had activity in the pouch region of the wing and haltere discs, as well as in the proximal region of leg discs (Fig. 7F). *Am_Ubx_37p2* drove expression in specific portions of the wing and haltere discs (Fig. 7G). Although *Ubx* is not expressed in the *D. melanogaster* wing disc (i.e., the forewing of the fly), wing activity has been observed previously

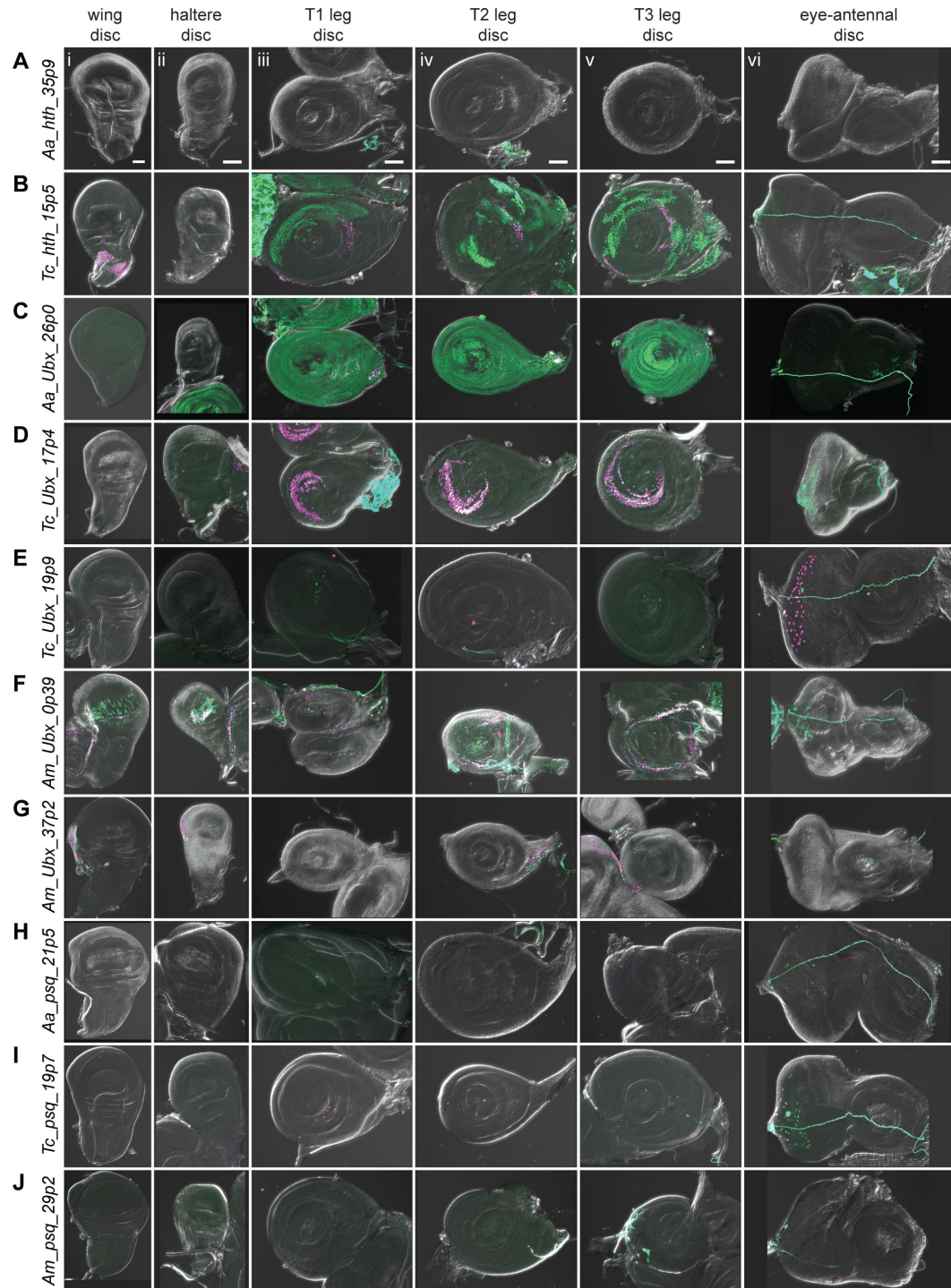


Figure 7

Reporter gene expression for tested *hth*, *Ubx*, and *psq* predicted enhancer sequences.

Each row shows expression for the indicated construct in (i) wing discs, (ii) haltere discs, (iii) T1 (prothoracic) leg discs, (iv) T2 (mesothoracic) leg discs, (v) T3 (metathoracic) leg discs, and (vi) eye-antennal discs (with eye portion to the left). Enhancer activities were detected by the G-TRACE system; magenta represents direct enhancer activity detected by dsRed expression, while green indicates lineage-based GFP expression. Scale bar is 50 μ m for each column.

with tested *Ubx* enhancer fragments (e.g. 42). Moreover, *Ubx* is expressed in both the forewing and hindwing discs in honeybees, making the reporter expression we observed driven by the two predicted *A. mellifera* enhancers consistent with their potential native activities (46 [↗](#)).

psq

We chose *pipsqueak* (*psq*), a transcription factor involved in Polycomb group gene silencing (47 [↗](#)), and its orthologous loci as the final targets for enhancer validation. There were four predicted *psq* enhancers in *D. melanogaster*, one of which, *Dm_psq_25p6*, overlaps known enhancer *psq_GMR41E12* (Supplemental Fig. S3). This enhancer is located within the second *psq* intron and drives expression in all imaginal discs (Fig. 5J [↗](#))(42 [↗](#)). The *A. aegypti* genome had four predictions for the *psq* locus (Supplemental Fig. S3); we chose *Aa_psq_21p5*, located in the second intron and with the highest local SCRMshaw score, for validation (Supplemental Fig. S3). However, *Aa_psq_21p5* did not have observable imaginal disc expression (Fig. 7H [↗](#)).

A. T. castaneum had only two predictions for the *psq* locus, both within the third intron (Supplemental Fig. S3). *Tc_psq_19p7*, which was chosen for validation due to its higher score, drove expression in the eye and leg discs (Fig. 7I [↗](#)).

From the *A. mellifera* genome, we selected the only prediction for the *psq* locus, *Am_psq_29p2* (Supplemental Fig. S3). *Am_psq_29p2* reporter activity was negative in all tissues assayed (Fig. 7J [↗](#)).

Embryonic activity

Although our SCRMshaw predictions were targeted toward imaginal disc activity, activity was also observed in embryos for many of the reporter lines (Supplemental Fig. S5). Analysis of this activity was complicated by the fact that our reporter vectors, while not having any basal activity in imaginal discs (Supplemental Fig. S4, A-E), displayed reporter gene expression in several tissues including hemocytes, caudal visceral mesoderm, and the proventriculus, even in the absence of a putative enhancer sequence (Supplemental Fig. S5A-D). In those lines that had reporter gene expression in other tissues (Supplemental Fig. S5I-S, X, Y), most of that expression was not clearly associated with the expected endogenous expression of the predicted target gene, although the complex embryonic expression patterns of these genes makes a definitive assessment difficult. Moreover, for most of the species, we do not currently know the expression patterns of either the gene in its native species (as opposed to the expression of its *Drosophila* ortholog), or the expression patterns of other nearby potential target genes. Further analysis, including additional control experiments, use of different reporter vectors, and assessment of gene expression patterns in each of the relevant species will be necessary before drawing final conclusions as to embryonic enhancer activity.

An insect regulatory annotation resource

Taken together, the results from our simulations, our comparisons to open chromatin regions, and our *in vivo* validation experiments demonstrate that SCRMshaw is remarkably effective at predicting regulatory sequences across a wide range of insect species. To facilitate access to our SCRMshaw-based regulatory annotations, we created a database with the results from our predictions using all training sets and all completed species. This database, which is freely accessible as part of the REDfly insect regulatory annotation site (16 [↗](#)), contains processed final prediction data and can be searched and filtered by gene, *D. melanogaster* ortholog, training set, enhancer location, and various other criteria. We will continue to add to this database as additional species and training sets are run through our annotation pipeline.

Discussion

The resource we introduce here is part of an ongoing effort to provide an initial regulatory annotation for all sequenced insects. These annotations are not complete, as we currently lack well-curated training sets for many tissues, including key embryonic tissues such as the central nervous system and many non-embryonic tissues. We aim to generate the necessary additional training sets over time, and add these new annotations to those presented here, along with comprehensive annotation for additional species. As a predictive pipeline, SCRMshaw is subject to the usual tradeoffs of sensitivity versus specificity in generating results. Sensitivity is difficult to assess, as there is no “complete” known set of enhancers for any organism, and it is currently impossible to make an accurate estimate of what the yield should be for any of our training models. Similarly, without comprehensive *in vivo* testing using multiple conditions and methods, an accurate false-positive rate cannot be computed. However, we presented here several lines of evidence demonstrating that true positives significantly outweigh false positives: we non-randomly predict multiple enhancers for specific subsets of loci; we predict enhancers in orthologous loci at a rate significantly higher than random expectation; our predictions are highly enriched for regions of accessible chromatin; and we achieve a high rate of validation using *in vivo* reporter gene assays.

Validation success rates

The results from the *in vivo* validation experiments are summarized in [Table 5](#). Overall, 17/22 (77%) of tested sequences revealed imaginal disc activity, consistent with our previous SCRMshaw success rates. 59% of these (10/17), or 45% of the total (10/22), had the correct target specificity of wing and/or haltere discs. This is again consistent with previous SCRMshaw experience, which shows that functional enhancers are predicted at a higher success rate than enhancers with specific targeted activity ([13](#), [14](#)).

These numbers may in fact underestimate the success rate of our predictions, for several reasons. One, all of the testing was performed in transgenic *D. melanogaster*, despite the putative enhancers being from three additional species. Reduced efficiency of certain transcription factor or cofactor binding, reduced enhancer-promoter compatibility, or other species-specific differences may lead to elevated false negative results. Two, predictions for *ex*, *klu*, and *ush* were made using a less-effective SCRMshaw post-processing algorithm and, for *A. mellifera*, a less-complete genome build. This may have led to suboptimal candidate enhancer selection. Three, it is possible that some of our predicted enhancers act as silencers rather than enhancers. Silencers, which attenuate rather than promote gene expression, are less well understood than enhancers, but in at least some instances have identical sequence characteristics (and in fact can act simultaneously as enhancers in some tissues and silencers in others) ([48](#), [49](#)). Our reporter gene assay was not designed to detect silencers, which would therefore appear as false-positive predictions. Indeed, an intriguing possibility is that some of our sequences that had reporter gene activity in non-targeted discs (e.g., leg discs) act as enhancers in those tissues but as silencers in the targeted wing discs. Alternatively, identified enhancers that drove expression in multiple imaginal discs may simply represent pleiotropic enhancers. Enhancer pleiotropy is not uncommon ([50](#), [51](#)), and moreover, there is overlap in the enhancer content of our training sets for the various discs. There are many gene expression and developmental similarities among the discs, and likely significant regulatory overlap; chromatin profiling experiments have found that wing, leg, and eye discs share the majority of their regions of accessible chromatin ([18](#), [32](#)).

Additional experiments will be necessary to distinguish between pleiotropic enhancers, dual-function silencers/enhancers, and other possibilities.

	targeted disc				non-targeted disc			
<i>ex</i>								
<i>klu</i>								
<i>ush</i>								
<i>hth</i>								
<i>Ubx</i>								
<i>Ubx</i>								
<i>psq</i>								

Results are shown for *D. melanogaster*, *T. castaneum*, *A. mellifera*, and *A. aegypti*. Black, positive for expression; grey, negative for expression; blue, expression observed in pupae but not in larvae.

Table 5

Summary of *in vivo* validation results

Redundant enhancers

Almost all of our training sets predicted multiple enhancers in certain loci at rates exceeding random expectation, consistent with the noted prevalence of redundant or “shadow” enhancers in numerous plant and animal species (reviewed by 27). The few training sets that did not may reflect types of tissues or regulatory processes for which having shadow enhancers is unusual, or may be indicative of poorly-performing training sets. Indeed, the sole training set that did not suggest the presence of shadow enhancers in either mosquito species, *adult_PNS*, also performed poorly by other metrics, such as failing to produce predictions in common over multiple species and scoring “poor” using our *pCRM_eval* training set evaluation tool (21 [↗](#)).

Although the evolutionary origins of shadow enhancers are not well understood, several studies suggest that at least one important contribution of shadow enhancers is to provide phenotypic robustness during development, particularly during stress conditions (52 [↗](#)–57 [↗](#)). In some cases, a mechanistic basis for this robustness has been suggested by the finding that different groups of transcription factors act on individual members of a shadow enhancer set (58 [↗](#), 59 [↗](#)). Further support for this idea can be found in the fact that shadow enhancers appear to have limited sequence similarity (58 [↗](#), 60 [↗](#)). However, our successes with SCRMshaw have demonstrated that enhancers can have little overt sequence similarity but still rely on the same underlying subsequence (and presumably, transcription factor binding) model. As the potential shadow enhancers we identify are predicted using the same training set, they are likely to be functioning using similar, rather than independent, regulatory mechanisms. Thus, shadow enhancers appear to come in at least two flavors: those that use similar mechanisms, such as those identified here, and those that use different mechanisms, such as the *Krüppel* enhancers studied by (58 [↗](#)).

Enhancer evolution at large divergence distances

SCRMshaw’s ability to find putatively “orthologous” enhancers—i.e., enhancers in orthologous loci predicted using the same regulatory model—not only substantiates the non-random nature of our prediction approach, but also opens the door to exciting studies of enhancer evolution over large divergence ranges. We previously illustrated the power of such an approach with an analysis of a small number of enhancers in just 2-3 highly diverged species (17 [↗](#)). However, the greatly increased number of species for which we now have regulatory predictions should allow us to follow the evolution of specific enhancers over their entire divergence range. Although the number of common loci tails off rapidly as the number of species considered increases, we suspect that this small (albeit statistically significant) number understates the true results. For one, only a limited number of species have been evaluated for common predictions so far, and these are not evenly distributed along the phylogenetic spectrum of sequenced species. Also, our analysis depends wholly on the presence of recognized *D. melanogaster* orthologous genes to define the common loci. This in turn is dependent on several factors, including the sensitivity and accuracy of our ortholog-calling pipeline, the reliability of the protein annotations we use in that pipeline, and on accurate target gene assignments. Each of these has known sources of error. For example, our ortholog-calling pipeline currently does not disambiguate multiple paralogs (see Methods), and ortholog detection has been shown to be sensitive to the method used for generating protein annotation, in particular when different annotation methods have been applied to different species in the comparison (61 [↗](#)). Moreover, our target gene assignments are presently based solely on closest-gene relationships, a method which is known to mis-assign a fair number of enhancer-gene target pairings (e.g. 62, 63, 64) and which does not take into account complexities in genome architecture such as nested genes, long non-coding RNAs, promoter competition, and the like (which are common in *D. melanogaster*, e.g. 65, 66). Developing effective and scalable solutions to these issues will be an important goal for future work.

As the number of species in our prediction database with fully mapped orthologs grows, it will become possible to ask increasingly sophisticated questions about the nature of enhancer evolution. For instance, we will be able to determine whether the numbers of putatively orthologous enhancer predictions follow phylogenetic relationships and degree of sequence divergence, and whether certain loci are only found in common for specific groups of species. Also in need of further investigation will be to determine whether sets of common enhancers are restricted to certain functional Gene Ontology categories, and how this might vary with phylogenetic grouping.

Ongoing regulatory annotation

While effective, SCRMshaw still has various limitations and aspects in need of improvement. Errors in genome assembly and insufficient repeat masking both appear to contribute to overly long predictions (multiple kilobases) that are unlikely to represent individual single enhancers (HA and MSH, unpublished observations). Poor assembly, while not having major effects on SCRMshaw overall, can significantly affect results at specific loci, as can inaccurate gene annotation (21 [↗](#)). Also requiring further investigation is how to best combine and weight the scores from the three individual SCRMshaw scoring methods of IMM, hexMCD, and PAC-rc. Individual SCRMshaw predictions should therefore be treated as just that—predictions—and appropriate validation experiments are recommended for any sequences of interest. The regulatory annotations presented here represent initial “1.0” versions of the regulatory genome. Like all genome annotations, these will continue to be revised as updated models become available, including addition of new training sets, confidence scores based on validation experiments, and improved genome builds. Nevertheless, the enrichment scores from our *in silico* experiments and the true-positive rates from our *in vivo* reporter gene experiments suggest that overall, true positive rates for most training sets are between 50-85%, with most likely having success rates exceeding 70%. Note that even this lower bound of a 50% true positive rate means that one out of every two predictions is correct—a more than acceptable rate to encourage follow-up experiments for predicted enhancers in organisms of interest. The catalog of predicted enhancers we introduce here, spanning 33 insect species and growing, should thus be useful resource for both large-scale and small-scale studies of the insect regulatory genome.

Methods

Datasets

Sequence and annotation for *D. melanogaster* were obtained from FlyBase (67 [↗](#)). For other species, wherever possible, genome sequence (FASTA) and annotation (GFF) files were downloaded from NCBI at www.ncbi.nlm.nih.gov/datasets [↗](#). Otherwise, the genome sequences and annotations were obtained from the primary literature or directly from the data generators (see **Table 1** [↗](#) for references). Version numbers for genomes and annotations are provided in **Table 1** [↗](#).

SCRMshaw

A detailed SCRMshaw protocol can be found at Asma et al., 2024 (*in preparation*). Genome files were masked using Tandem Repeat Finder (22 [↗](#)) using parameters 2 7 7 80 10 50 500 -m -h. Genome and annotation files were then assessed using our *preflight* script. Any sequence scaffolds not containing annotated genes were removed before passing the genome sequence to the main SCRMshaw program. SCRMshaw was run using the “HD” default settings as described in Asma and Halfon (21 [↗](#)). SCRMshaw can be downloaded at https://github.com/HalfonLab/SCRMshaw_HD [↗](#).

Postprocessing

The top 5000 hits were extracted using scripts *Generate_top_N_SCRM-hits.pl* and *concatenatingOffsetResults.sh* and passed to *postProcessingScrmshawPipeline.py* with *-num* = “5000” and *-topN* = “Median.” Post-processing was performed essentially as previously described (15), with the following modifications (Supplemental Fig. S6): before evaluating each 10 bp region, scores from each of the 25 individual SCRMshaw instances were assessed and any 500 bp window whose score was below the value of the 5000th ranked score was eliminated by having its score reset to zero (Supplemental Fig. S6B). The “elbow” point of the SCRMshaw score curve of the 5000 top scores from each instance was then determined (Fig. S6C), and any scores below the elbow point were reset to zero (Supplemental Fig. S6D; gray boxes). Only after these two rounds of score evaluation were all windows grouped together (Fig. S6E, F) and subjected to peak calling on 10 bp intervals (Fig. S6G). Final “top predictions” were then any peaks with an amplitude above the selected amplitude threshold (elbow point of amplitude curve, represented by a red dot in Supplemental Fig. S6H), following the peak-calling step. See (21) and Asma et al. 2024 (*in preparation*) for details.

All scripts are available at <https://github.com/HalfonLab/>.

Orthology mapping

The final SCRMshaw output from the above steps was used as input to our orthology mapping pipeline. For each species, a FASTA-formatted file of all annotated proteins was downloaded from NCBI (www.ncbi.nlm.nih.gov/datasets). The annotated proteins from *D. melanogaster* plus each individual other species were used as input to *Orthologer* (24) to obtain the *Drosophila* ortholog for each protein (when existing). Our approach was designed to be minimally restrictive in that we did not enforce a one-to-one ortholog mapping; in cases of likely paralogs, we considered all of the paralogs as a potential result. Details on the orthology mapping protocol can be found in Asma et al. 2024 (*in preparation*).

Evaluating the number of predictions per locus

For each training set, BEDTools “merge” was used to remove any overlapping predictions (68). The results were then permuted 1000 times using BEDTools “shuffle”, with coding regions excluded. BEDTools “closest” was used to assign upstream and downstream flanking genes to each of the permuted predictions, with the following parameters: the ‘-io’ flag was enabled to ignore any overlaps between predictions and genes, and ‘-D ref -id’ and ‘-D ref -iu’ were used to obtain the closest 5’ and 3’ genes (with respect to chromosome coordinates), respectively. A custom Python script, *checkSameLocus_ForSimulations.py* (located at https://github.com/HalfonLab/UtilityPrograms/blob/master/checkSameLocus_ForSimulations.py) was used to calculate the number of predictions per locus for both the real and permuted results. For this purpose, “locus” was defined as the entire region between the left and right flanking genes for each SCRMshaw prediction; that is, for each prediction we take the region between the 3’ end of the upstream gene and the transcription start site of the downstream gene. If a prediction is within an intron, we define the locus as the entire span of the enclosing gene. If a prediction overlaps two genes, the locus is considered to be the entire span of the two genes.

Average intergenic region sizes for the genomes were estimated using the distances between neighboring genes, discarding any nested or overlapping gene pairs, as provided by the SCRMshaw *preflight* script.

Significance was assessed by calculating the empirical *P*-value, defined as the number of real loci with a number of predictions greater than the maximum number obtained from the 1000 simulations, for each locus containing at least one SCRMshaw prediction.

Evaluating the number of predictions in common across species

To determine the expected number of common predictions—i.e., predictions in orthologous loci—across species, we utilized SCRMshaw predictions for 16 species. Each set of predictions was sorted, merged, permuted, and mapped to new loci as described above for “Evaluating the number of predictions per locus.” The permuted predictions were used as input for the script “*checkSameLocus_ForSimulations_crossSpecies.py*” (located at https://github.com/HalfonLab/UtilityPrograms/blob/master/checkSameLocus_ForSimulations_crossSpecies.py). This script identifies the *Drosophila* orthologs of the nearest flanking genes for each species and calculates the number of common loci flanking the simulated SCRMshaw predictions for 5, 10, 11, 12, 13, 14, 15, and 16 species. For each species, the process was repeated for a total of 360 permutations. The mean and standard deviation of the permuted results were then used to calculate a z-score for each training set. We considered training sets with z-score ≥ 1.645 to be significant ($P < 0.05$, not corrected for multiple testing).

Overlap between SCRMshaw predictions and open chromatin regions

Open chromatin data were obtained from the following sources:

1. *D. melanogaster* : Data for *D. melanogaster* were downloaded from GEO and consisted of ATAC-seq and FAIRE-seq data from accessions GSE101827 , GSE38727 and GSE118240. These assays were performed using blastoderm embryos, eye-antennal discs, wing discs, haltere discs, leg discs, third instar central nervous system, and wing, leg, and haltere pharate appendages (31 [↗](#), 32 [↗](#), 69 [↗](#)).
2. *T. castaneum*: FAIRE-seq data for three stages of embryogenesis, larval central nervous system, and larval second and third thoracic epidermal tissues were downloaded from GEO (GSE104495)(18 [↗](#)). The FAIRE profiles were remapped to the version 5.2 of the *T. castaneum* genome (Tcas5.2) for this study. The remapped FAIRE profiles are available on iBeetle-Base (<https://ibeetle-base.uni-goettingen.de/>) [↗](#) (70 [↗](#)). *A. gambiae*: ATAC-seq data for adult midgut and salivary gland were downloaded from GEO (GSE152924)(71 [↗](#)).
3. *D. plexippus*, *J. coenia*, *H. himera*, and *V. cardui*: ATAC-seq data for larval forewing and hindwing tissues at stage M5 were provided by Anyi Mazo-Vargas and Robert Reed (33 [↗](#)).

The overlap between SCRMshaw predictions and open chromatin regions was determined using BEDTools “intersect” with parameters -wa -u -f 0.1 such that sequences needed to overlap at least 10% of their length and were considered a single overlap in the event that more than one open chromatin peak overlapped a prediction.

To assess significance, the SCRMshaw predictions were permuted 500 times using BEDTools “shuffle” and the overlaps with open chromatin regions assessed as above. The mean and standard deviation of the permuted results were then used to calculate a z-score for each training set.

We also calculated a “fold enrichment” score by dividing the observed number of overlapping regions by the expected number, to provide a sense of effect size in addition to statistical significance.

Reporter constructs and transgenic *Drosophila*

Sequences for reporter gene analysis, including attL1 and attL2 sites, were synthesized *de novo* and cloned into pUC57 Kan-r (GenScript, Piscataway NJ) as entry vectors suitable for Gateway cloning (72 [↗](#)). Gateway LR recombination was then used to move the sequences into piggyPhiGUGd (*hth*, *Ubx*, and *psq* lines) and piggyPhiGUGd-TomatoI (*ex*, *klu*, and *ush* lines) (Deem

et al., *submitted*). Transgenic flies were generated by BestGene (Chino Hills, CA) using PhiC31 recombination and the attP2 third chromosome insertion site. piggyPhiGUGd lines were subsequently crossed to G-TRACE for visualizing enhancer activities (34 [↗](#)).

For each construct, imaginal discs from at least six larvae were dissected and mounted for direct fluorescence visualization using a Zeiss Axio Imager M2 microscope with ApoTome 2. Embryos were fixed and stained using standard *Drosophila* methods using anti-dsRed (Clontech) and visualized using the ABC-HRP kit (VectorLabs, Newark CA).

Database implementation

The SCRMshaw results database is implemented as part of REDfly (16 [↗](#)), a MariaDB-based database hosted on a private OpenStack cloud infrastructure maintained by the University at Buffalo Center for Computational Research. As part of an ongoing transition of REDfly to a more modern software architecture, backend functions are implemented in Node.JS and Python, while frontend components utilize React.JS and Next.js. GraphQL is used as the query language. REDfly is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives License v4 International (CC BY-NC-ND 4.0) and its underlying source code under a GNU General Public License v3 (GNU GPL 3.0).

Image credits

The insect silhouettes in **Table 5** [↗](#) were obtained from The Noun Project (thenounproject.com [↗](#)), artist Georgiana Ionescu, under a Creative Commons CC-BY 3.0 license.

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Supporting information

S1 Text: Legends for Supplemental Figures

S1 Figure: Lengths of predicted enhancers, including long outliers

S2 Figure: Open chromatin data for predictions in the *ex*, *klu*, and *ush* loci

S3 Figure: Open chromatin data for predictions in the *hth*, *Ubx*, and *psq* loci

S4 Figure: Additional expression observed in selected transgenic reporter lines

S5 Figure: Reporter gene expression observed in embryos

S6 Figure: Revised post-processing method used for SCRMshaw

S1 Table: SCRMshaw training sets used in this study

S2 Table: Number of predicted enhancers for each species, by method

S3 Table: Real and simulated predictions per locus

S4 Table: Real and simulated counts of predictions in orthologous loci

S5 Table: Primers and sequences used for *in vivo* validation

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Reviewer #1 (Public Review):

Summary:

The authors provide a genome annotation resource of 33 insects using a motif-blind prediction method for tissue-specific cis-regulatory modules. This is a welcome addition that may facilitate further research in new laboratory systems, and the approach seems to be relatively accurate, although it should be combined with other sources of evidence to be practical.

Strengths:

The paper clearly presents the resource, including the testing of candidate enhancers identified from various insects in *Drosophila*. This cross-species analysis, and the inherent suggestion that training datasets generated in flies can predict a cis-regulatory activity in distant insects, is interesting. While I can not be sure this approach will prevail in the future, for example with approaches that leverage the prediction of TF binding motifs, the SCRMSHaw tool is certainly useful and worth consideration for the large community of genome scientists working on insects.

Weaknesses:

While the authors made the effort to provide access to the SCRMSHaw annotations via the RedFly database, the usefulness of this resource is somewhat limited at the moment. First, it is possible to generate tables of annotated elements with coordinates, but it would be more useful to allow downloads of the 33 genome annotations in GFF (or equivalent) format, with SCRMSHaw predictions appearing as a new feature. Also, I should note that unlike most species some annotations seem to have issues in the current RedFly implementation. For example, Vcar and Jcoen turn empty.

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Reviewer #2 (Public Review):

Summary:

The ability of researchers to identify and compare enhancers across different species is an important facet of understanding gene regulation across development and evolution. Many traditional methods of enhancer identification involve sequence alignments and manual annotations, limiting the ability to expand the scope of regulatory investigations into many species. In order to overcome this obstacle, the authors apply a previously published machine learning method called SCRMSHaw to predict enhancers across 33 insect species, using *D. melanogaster* as a reference. SCRMSHaw operates through the selection of a few dozen training loci in a reference genome, marking genomic loci in other species that are significantly enriched with similar k-mer distributions relative to randomly selected genomic backgrounds. Upon identification of predicted enhancer regions, the authors perform post-processing step filtering and identify the most likely predicted enhancer candidates based on the proximity of an orthologous target gene. They then perform reporter gene analysis to validate selected predicted enhancers from other species in *D. melanogaster*. The analysis of the expression patterns returned variable results across the selected predicted regions.

Strengths:

The authors provide annotations of predicted regions across dozens of insect species, with the intention of expanding and refining the annotations for use by the scientific field. This is useful, as researchers will be able to use the identified annotations for their own work or as a benchmark for future methods. This work also showcases the flexible and versatile nature of SCRMSHaw, which can readily obtain predictions using training sets of genomic loci requiring only a few dozen annotations as input. SCRMSHaw does not require sequence alignments of the enhancers and can operate without prior knowledge of the cis-regulatory sequence rules such as transcription factor binding motifs, making it a useful tool to explore the evolution of enhancers in further distant and less well-studied species.

Weaknesses:

This work provides predicted enhancer annotations across many insect species, with reporter gene analysis being conducted on selected regions to test the predictions. However, the code

for the SCRMshaw analysis pipeline used in this work is not made available, making reproducibility of this work difficult. Additionally, while the authors claim the predicted enhancers are available within the REDfly database, the predicted enhancer coordinates are currently not downloadable as Supplementary Material or from a linked resource.

The authors do not validate or benchmark the application of SCRMshaw against other published methods, nor do they seek to apply SCRMshaw under a variety of conditions to confirm the robustness of the returned predicted enhancers across species. Since SCRMshaw relies on an established k-mer enrichment of the training loci, its performance is presumably highly sensitive to the selection of training regions as well as the statistical power of the given k-mer counts. The authors do not justify their selection of training regions by which they perform predictions.

While there is an attempt made to report and validate the annotated predicted enhancers using previously published data and tools, the validation lacks the depth to conclude with confidence that the predicted set of regions across each species is of high quality. In vivo, reporter assays were conducted to anecdotally confirm the validity of a few selected regions experimentally, but even these results are difficult to interpret. There is no large-scale attempt to assess the conservation of enhancer function across all annotated species.

Lastly, it is suggested that predicted regions are derived from the shared presence of sequence features such as transcription factor binding motifs, detected through k-mer enrichment via SCRMshaw. This assumption has not been examined, although there are public motif discovery tools that would be appropriate to discover whether SCRMshaw is assigning predicted regions based on previously understood motif grammar, or due to other sequence patterns captured by k-mer count distributions. Understanding the sequence-derived nature of what drives predictions is within the scope of this work and would boost confidence in the predicted enhancers, even if it is limited to a few training examples for the sake of clarity of interpretation.

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Reviewer #3 (Public Review):

Summary:

In this ambitious paper, the authors develop an unparalleled community resource of insect genome regulatory annotations spanning five insect orders. They employ their previously-developed SCRMshaw method for computational cross-species enhancer prediction, drawing on available training datasets of validated enhancer sequence and expression from *Drosophila melanogaster*, which had been previously shown to perform well across select holometabolous insects (representing 160-345MY divergence). In this work, they expand regulatory sequence annotation to 33 insect genomes spanning Holometabola and Hemiptera, which is even more distantly related to the fly model. They perform multiple downstream analyses of sets of predicted enhancers to assess the true-positive rate of predictions; the independent comparisons of real predictions with simulated predictions and with chromatin accessibility data, as well as the functional validation through reporter gene analysis, strengthen their conclusions that their annotation pipeline achieves a high true-positive rate and can be used across long divergence times to computationally annotate regulatory genome regions, an ability that has been previously inaccessible for non-model insects and now is possible across the many newly-sequenced insect scaffold-level genomes.

Strengths:

This work fills a large gap in current methods and resources for predicting regulatory regions of the genome, a task that has long lagged behind that of coding region prediction and analysis.

Despite technical constraints in working outside of well-developed model insect systems, the authors creatively draw on existing resources to scaffold a pipeline and independently assess the likelihood of prediction validity.

The established database will be a welcome community resource in its current state, and even more so as the authors continue to expand their annotations to more insect genomes as they indicate. Their available analysis pipeline itself will be useful to the community as well for research groups that may want to undertake their own regulatory genome annotation.

Weaknesses:

The rates of predicted true positive enhancer identification vary widely across the genomes included here based on the simulations and comparison to datasets of accessible chromatin in a manner that doesn't map neatly onto phylogenetic distance. At this point, it is unclear why these patterns may arise, although this may become more clear as regulatory annotation is undertaken for more genomes.

Functional assessment of predicted enhancers was performed through reporter gene assays primarily in *Drosophila melanogaster* imaginal discs, a system amenable to transgenics. Unfortunately, this mode of canonical imaginal disc development is only representative of a subset of all holometabolous insects; therefore, it is difficult to interpret reporter gene expression in a fly imaginal disc as evidence of a true positive enhancer that would be active in its native species whose adult appendages develop differently through the larval stage (for example, Coleopteran and Lepidopteran legs). However, the reporter gene assays from other tissues do offer strong evidence of true positive enhancer detection, and constraints on transgenic experiments in other systems mean that this approach is the best available.

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Author response:

We thank the reviewers for their thoughtful and insightful comments. We were pleased to see that the reviewers and editors consider our work a “welcome addition” that “fills a large gap” in comparative genomics methods and provides “an unparalleled community resource of insect genome regulatory annotations.”

Many of the reviewers' comments reflect weaknesses in our description of the methodology. As the basic SCRMshaw methodology has been published previously, we had opted for brevity over detail in the current manuscript. We recognize now that we went too far in that direction, and we will include more methodological detail in our revised submission, along with easier access to the code we used. The reviewers also offered some helpful suggestions regarding data availability which we intend to address, including direct download of the results in GFF format and adding to the results database several species that were inadvertently omitted.

Reviewer 2 expressed concerns about benchmarking SCRMshaw against other methods. We respectfully feel this lies outside the scope of the current study, which focuses on application of SCRMshaw to generate a multi-species annotation resource rather than on an attempt to show that SCRMshaw is superior to other approaches. We provide evidence in this manuscript, as well as in previous publications, that supports the effectiveness of SCRMshaw as an approach for regulatory element discovery that is suitable for the task at hand.

Benchmarking for regulatory element discovery brings many challenges, as there are no comprehensive “truth” sets to serve as a comparison baseline. We therefore do not attempt strong claims here about the relative merits of SCRMshaw vs. other methods (although we have explored this in previous publications). Note that we also previously demonstrated commonality of transcription factor binding sites in cross-species SCRMshaw predictions, in particular in Kazemian et al. 2014 (*Genome Biol. Evol.* 6:2301).

Finally, because it has important implications for understanding our results, we would like to point out a small misconception in Reviewer 2’s Summary of our study. The reviewer states that we “identify the most likely predicted enhancer candidates based on the proximity of an orthologous target gene.” We stress, however, that putative target gene assignments and identities have no impact at all on our prediction of regulatory sequences. Predictions are solely based on sequence-dependent SCRMshaw scores, with no regard to the nature or identities of nearby annotated features. Putative target genes are mapped to *Drosophila* orthologs purely as a convenience to aid in interpreting and prioritizing the predicted regulatory elements. We will take care to clarify this important point in our revised submission.

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